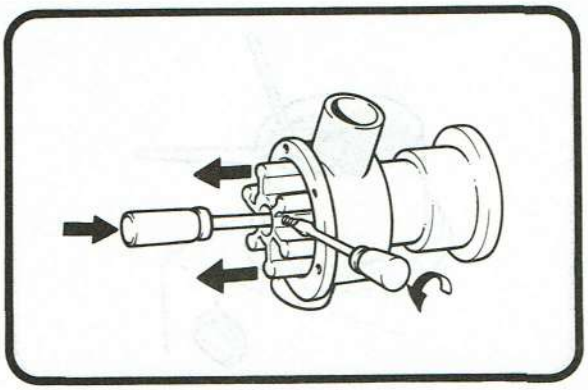


Figure 5-2. *Some flexible impellers also are secured to the shaft by a set screw. You must pry these out about 1/2 inch to undo the set screw. (Courtesy Volvo Penta)*

Overheating During Normal Operation.

Check the oil level. If a low oil level is causing the engine to overheat, expensive damage may be in the making. A partial seizure of one or more pistons may already be generating a tremendous amount of heat on the cylinder walls. More likely than a lack of oil, however, is a reduction in the raw-water flow through the cooling system.

The raw-water inlet screen (if fitted) on the outside of the hull may be blocked with a piece of plastic. Throttle down, put the boat in reverse, and throttle up. With any luck, the reverse propeller thrust will wash it clear. To confirm this, loosen a hose below the waterline and see if water flows into the boat. If the flow is restricted, check also for barnacles on the inlet

Troubleshooting Chart 5-2.
Overheating in Operation.

Check the raw-water overboard discharge. Is the flow less than normal? NO	YES Check for obstructions in the raw-water circuit (see troubleshooting chart 5-1). In addition, check the raw-water circuit for silting, scale and other partial obstructions (see the text).
Check the oil level. Is it low? NO	YES Refill with the correct grade and viscosity of oil.
Is the boat overloaded? (Check for a rope around the propeller; a heavily fouled bottom; adverse conditions; excessive auxiliary equipment; an oversized propeller.) NO	YES Reduce the loading.
Check the coolant level in the fresh-water circuit (if fitted). Caution: do not remove the header-tank pressure-cap when hot. Is the level low? NO	YES Refill and find the leak.
Is the fresh-water circuit air-locked? (With the header-tank pressure-cap off, check for signs of flow when the engine is running—you may have to wait for the engine to cool down and the thermostat to close.) NO	YES Break the hoses loose at the fresh-water pump and water heater (if fitted) and bleed off any air.
Is the thermostat operating incorrectly? (Remove and test as outlined in the text.) NO	YES Replace.
There is probably a mechanical problem (e.g. faulty fuel injection; a partial seizure)—see the text.	